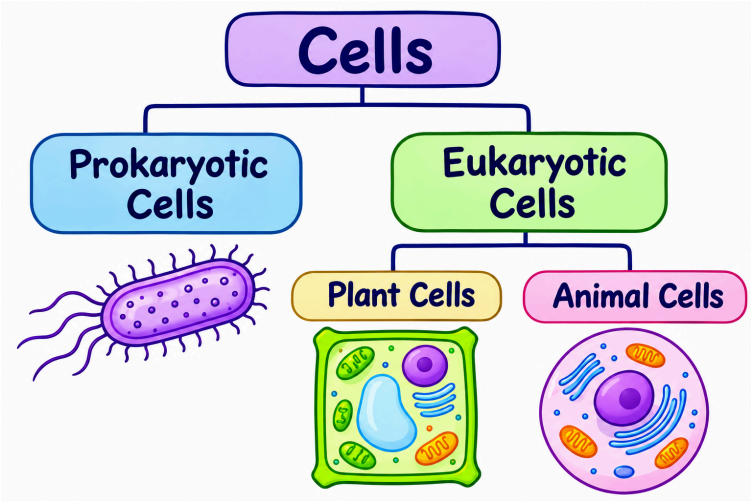
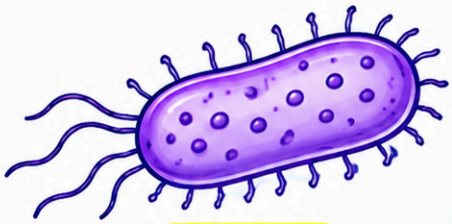


- Cells are of two main types based on the presence or absence of a true nucleus.

(A) PROKARYOTIC CELLS

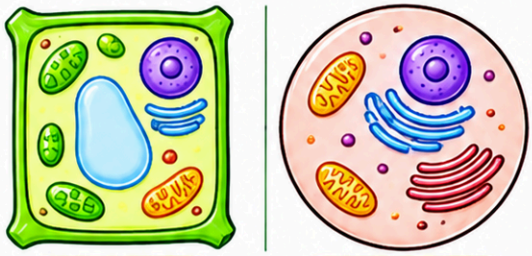
- Do not have a true nucleus.
- Do not have membrane-bound organelles.
- Usually smaller and simpler.
- Examples: Bacteria (e.g., E. coli).



Bacteria

(B) EUKARYOTIC CELLS

- Have a true nucleus.
- Contain membrane-bound organelles (e.g., mitochondria, ER, Golgi apparatus).
- Usually larger and more complex.
- Examples: Plant cells and animal cells.



Plant cell | Animal cell

Key Difference

Prokaryotic Cells	Eukaryotic Cells
No true nucleus.	Have a true nucleus.
No membrane-bound organelles.	Contain membrane-bound organelles.
Smaller and simpler (e.g., bacteria).	Larger and more complex (e.g., plant and animal cells).

PLANT CELL

- A plant cell is a eukaryotic cell.

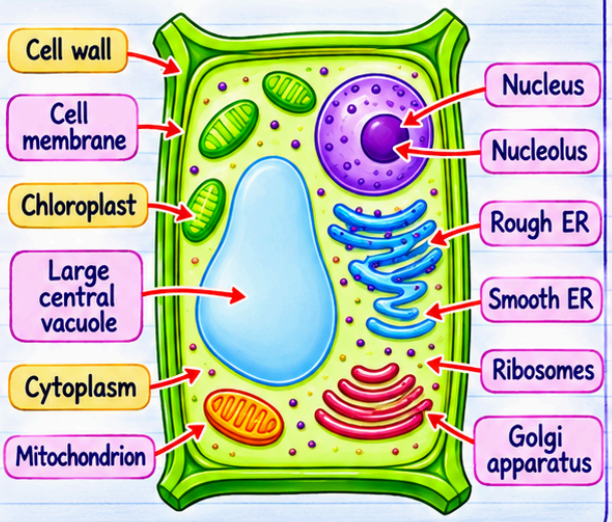
MAIN FEATURES :

- Has a cell wall.
- Has a cell membrane.
- Contains chloroplasts.
- Usually has a large central vacuole.
- Has a nucleus and cytoplasm.

IMPORTANT :

- Chloroplasts help plants make food by photosynthesis.
- The cell wall provides support and protection.

PLANT CELL (Labeled)



ANIMAL CELL

- An animal cell is also a eukaryotic cell.

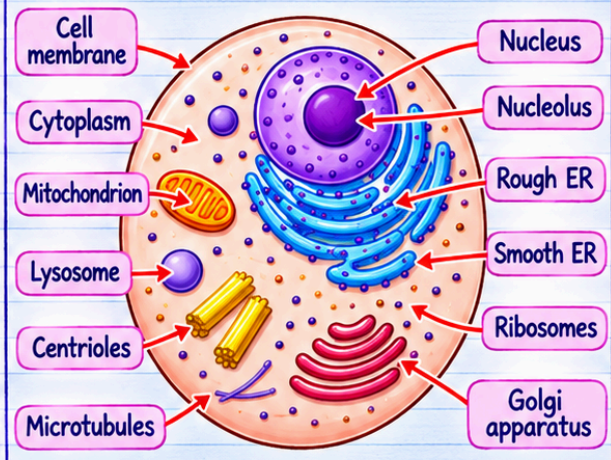
MAIN FEATURES :

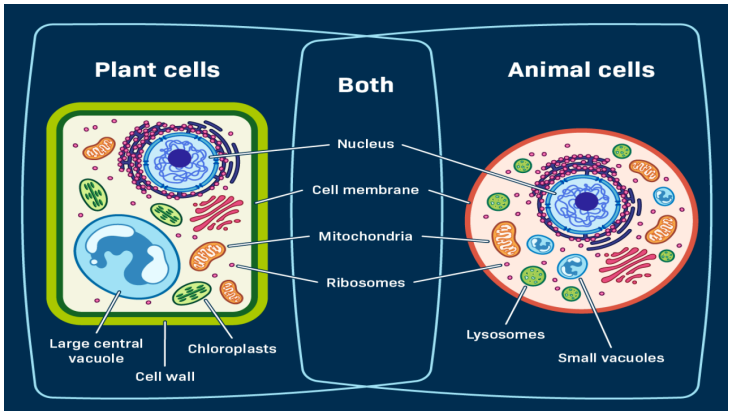
- Has a cell membrane.
- Has a nucleus and cytoplasm.
- Contains mitochondria.
- Usually has small vacuoles.
- Has no cell wall.
- Has no chloroplasts.

IMPORTANT :

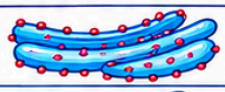
- Animal cells are usually round or irregular in shape.

ANIMAL CELL (Labeled)





Main Cell Organelles and Their Functions

Organelle	Function	Simple Diagram
Nucleus	Controls all cell activities.	
Cell membrane	Controls the entry and exit of substances into and out of the cell.	
Cytoplasm	Jelly-like substance where many cell reactions take place.	
Mitochondria	Release energy from food (site of respiration).	
Ribosomes	Make proteins.	
Rough ER	Helps make and transport proteins. (Has ribosomes on it.)	
Smooth ER	Makes lipids (fats) and helps in transport.	
Golgi apparatus	Modifies, packages and transports substances (such as proteins).	
Vacuole	Stores water and other materials (e.g., food, waste).	
Chloroplast	Makes food using sunlight (photosynthesis). (only in plant cells)	
Cell wall	Gives support, protection and shape to the cell. (only in plant cells)	
Lysosomes	Break down waste materials and worn-out cell parts. (more common in animal cells)	

Organelle	Function (in short)
Nucleus	Control centre of the cell.
Mitochondria	Release energy.
Ribosomes	Make proteins.
Chloroplast	Carry out photosynthesis (in plant cells).
Cell wall	Provide support and protection (in plant cells).
Vacuole	Storage of water and other materials.

Remember:

Different cell organelles work together to keep the cell alive.